# Exit Ticket · reflect and retrieve

## Exit Ticket

Passing an integer parameter by value means changing the parameter does not change the original integer variable.

## Use key vocabulary from memory

1. Explain Scope in your own words.

2. Explain Local variable in your own words.

3. Explain Global variable in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.





# Do Now · retrieval

## Scope & parameter passing in C#

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

What is the difference between a C# function and a procedure-style method?

### Q2

Which C# loop always executes its body at least once?

### Q3

What does == do in a C# condition?

### Q4

In Triple(7) , identify the argument. In Triple(int number) , identify the parameter.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Scope controls where a variable can be used — A local variable is declared inside a method or block and can only be used within that scope. — For this course, we use a class-level static field to demonstrate the OCR idea of a global variable that can be accessed by multiple methods.

## Core idea 2

Parameters are local to the method — A parameter receives data when a method is called and behaves as a local variable inside that method. — CODE:

```
public static int Triple(int number)
{
return number * 3;
}
```

## Core idea 3

By value = the method receives a copy — CODE:

```
public static void Change(int number)
{
number = 100;
}
```

— Changing number changes only the method's local copy. The original variable in Main() is unchanged.

## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 2

Swap two values
Create Swap(ref int first, ref int second) .
Use one local temporary variable so that the two original values exchange places.
In Main() , start with a = 10 and b = 25 , call the method and output both values.

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Local variables in two methods
Create two procedure-style methods: First() and Second() .
Inside each method declare a local variable called result , but give it a different value. Output each value.

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Core idea 4

ref gives access to the original variable — CODE:
public static void Change(ref int number)
{
number = 100;
}

// call
Change(ref value); — Write ref in both places: in the method parameter and in the method call.

## Vocabulary · use the correct technical terms

Scope — The part of a program in which a variable can be accessed.

Local variable — A variable visible only within the method or block where it is declared.

Global variable — A variable that can be accessed from multiple parts of a program.

Parameter — The named input in a method definition.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

Why doesn't the original value change?

## Follow the worked method

1. Start in Main — CODE:
int score = 10; — The variable score belongs to Main() and currently stores 10.

2. Only number becomes 100 — CODE:
ChangeByValue(score);

inside method:
number = 100

back in Main:
score = 10 — When the method ends, its local parameter is no longer needed.

3. Reference passing changes the original — CODE:
ChangeByReference(ref score);

```
// method
public static void ChangeByReference(ref int number)
{
number = 100;
}
```

## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Classify each variable

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.